MILITARY SPECIFICATION SHEET

ELECTRON TUBES, GAS SWITCHING

TYPES 5921 AND 5922

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: ATR, transmitter peak power output = 1,000 kw

ABSOLUTE RATINGS:

Parameter:	Transmitter po	TA	Alt
Unit:	kw	°C	ft
Maximum:	1,000	+100	10,000
Minimum:	20	-40	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any - See note 2

TEST CONDITIONS:

Parameter:	Transmitter po	tp	Du	Prr	VSWR	F
Unit:	kw	μs	---	pps	---	MHz
Test 1:						
Maximum:	---	---	---	---	---	---
:	20	---	0.001	---	---	F2
Minimum:	---	---	---	---	---	---
Test 2:						
Maximum:	---	1.10	---	1,100	1.05	---
:	20	1.00	---	1,000	---	F1
Minimum:	---	0.90	---	900	---	---
Test 3:						
Maximum:	---	---	---	---	1.20	---
:	---	---	0.001	---	---	F3
Minimum:	750	---	---	---	---	---

Frequency			
	MHz		
F	Type 5921	Type 5922	±
Fo	3,200	3,400	3.5
F1	3,300	3,300	33
F2	3,350	3,350	100
F3	3,425	3,425	25

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4021	Degradation due to vibration	---	---		---	---	---	---
4461	Loaded Q	2	---		QL	---	5.5	---
4474	High-level VSWR	---	2		---	---	1.15	---
	Quality conformance inspection, part 1	4	---					
4482	Tuning susceptance	2	---	F = Fo	b	-0.05	+0.05	---
4486	Firing time	2, 3	1		t	---	10	sec
4488	Arc loss	2, 3	1		La	---	0.8	dB
1027	Temperature cycling (nonoperating)	---	---		---	---	---	---
1027	Temperature cycling life test	---	---	Group C	---	50	---	Cycles
	Quality conformance inspection, part 2							
4101	Dielectric material strain	---	---		---	---	---	---
1036	Bump	1	---	Hammer angle 25 degrees	---	---	---	---
4484	Normalized conductance	2	---	F = Fo	g	---	0.05	---
	Quality conformance inspection, part 3							
4551	Life test	---	3	Group D	t	1,000	---	hrs
---	Life-test end points	---	---					
4482	Tuning susceptance	2	---	F = Fo	b	-0.05	+0.05	---
4484	Normalized conductance	2	---	F = Fo	g	---	0.1	---
4486	Firing time	2, 3	1		t	---	10	sec

NOTES:

1. The tube should not be hit directly with the hammer. An anvil as shown on Drawing 188-JAN may be used.

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NOTES: -Continued

2. Mount in accordance with Drawing 153-JAN.

3. The tube shall be mounted as in note 2 and followed by a matched load. The tube shall fire within the limits specified after application of rf power. This test shall be performed at least 7 days after pumping and at least 24 hours after any previous discharge.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

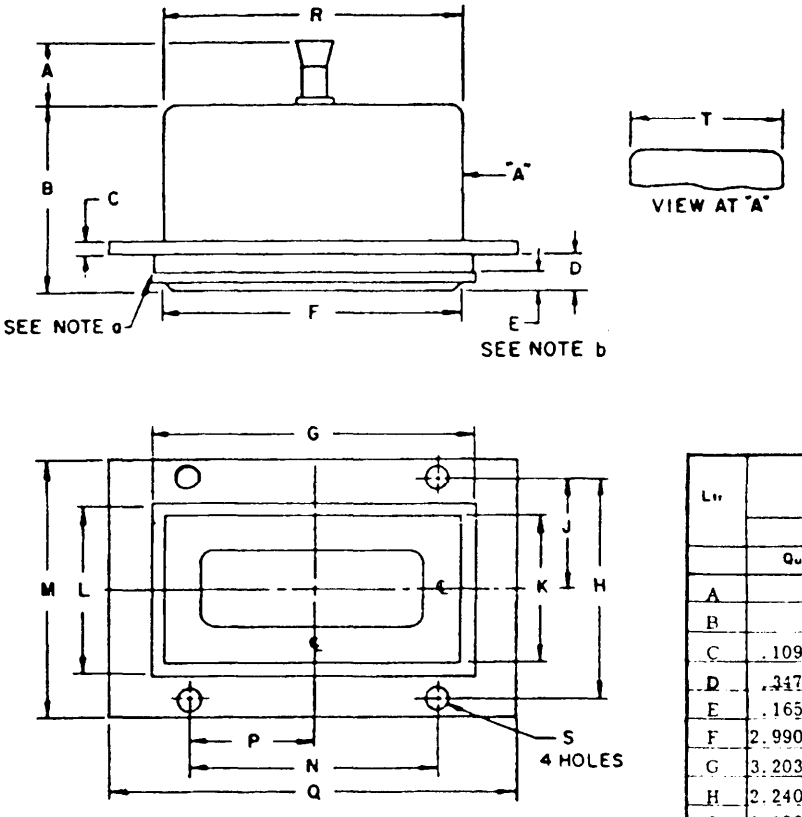
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-69)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, Part 1 (SEE NOTE 4)	
A		.625 (15.88)
B		1.875 (47.63)
C	.109 (2.77)	.141 (3.58)
D	.317 (8.81)	.377 (9.58)
E	.165 (4.19)	.195 (4.95)
F	2.990 (75.95)	3.010 (76.45)
G	3.203 (81.36)	3.233 (82.14)
H	2.240 (56.90)	2.260 (57.40)
J	1.120 (28.45)	1.130 (28.70)
K	1.490 (37.85)	1.510 (38.35)
L	1.703 (43.26)	1.733 (44.02)
M	2.609 (66.27)	2.641 (67.08)
N	2.490 (63.25)	2.510 (63.75)
P	1.245 (31.62)	1.255 (31.88)
Q	4.109 (104.37)	4.141 (105.18)
R	2.985 (75.82)	3.015 (76.58)
S	.203 (5.16) Dia	.234 (5.94) Dia
T	1.484 (37.69)	1.516 (38.51)

NOTES:

a. Gasket per drawing 189-JAN. Gasket to be firmly attached for a minimum of 75 percent of contact area.

b. Dimension E shall be measured prior to the attachment of the gasket to the tube.

FIGURE 1. Outline drawing of electron tube types 5921 and 5922.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25234

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